

Protocol	Full Form	Purpose	Default Port
HTTP	HyperText Transfer Protocol	Opens websites	80
HTTPS	HyperText Transfer Protocol Secure	Secure websites	443
FTP	File Transfer Protocol	Transfer files	21
SSH	Secure Shell	Remote login to Linux	22
Telnet	Telecommunication Network	Remote login (not secure)	23
SMTP	Simple Mail Transfer Protocol	Send email	25
DNS	Domain Name System	Converts website names to IP addresses	53
DHCP	Dynamic Host Configuration Protocol	Automatically assigns IP addresses	67/68
NFS	Network File System	Share files between Linux systems	2049

What is protocol

A protocol is a set of rules that tells computers how to send and receive data.

1.What is server?

A **server** is a computer or software that provides services, data, or resources to other computers, called **clients**, over a network.

Clients send requests.

Server gives responses.

Types of server

1.Web Server

Purpose: Hosts websites and web applications.

Examples:

- Apache
- Nginx

Interview answer:

“A web server stores website files and delivers web pages to users through a browser.”

2. File Server

Purpose: Stores and shares files over a network.

Interview answer:

“A file server allows multiple users to store, access, and share files from a central location.”

3. Mail Server

Purpose: Sends and receives emails.

Interview answer:

“A mail server manages email communication between users.”

4. FTP Server

Purpose: Transfers files between computers.

Interview answer:

“An FTP server is used to upload and download files over a network.”

Server Type	Purpose
Web Server	Hosts websites
File Server	Shares files
Database Server	Stores data
Mail Server	Sends and receives emails
DNS Server	Converts domain names to IP addresses
DHCP Server	Assigns IP addresses

Server Type	Purpose
FTP Server	Transfers files
NFS Server	Shares Linux directories
Print Server	Manages printers
Application Server	Runs applications

NFS

Network File System.

It is a protocol used to share files and directories between Linux systems over a network.

One Linux server shares files, many Linux clients use them."

#Package name : nfs-utils

#Service name : nfs-server.service

#Default port : 2049

We use NFS to:

- Share files between Linux systems.
- Save storage space.
- Manage files from one central location.
- Make collaboration easier.
- Simplify backups.

1. Install the NFS server package : In server side

#yum install nfs-utils

```
[root@server ~]# yum install nfs-utils
CentOS Stream 9 - BaseOS                               11 kB/s | 5.4 kB    00:00
CentOS Stream 9 - BaseOS                               666 kB/s | 8.9 MB   00:13
CentOS Stream 9 - AppStream                             7.0 kB/s | 5.5 kB    00:00
CentOS Stream 9 - AppStream                             882 kB/s | 28 MB    00:32
CentOS Stream 9 - Extras packages                       10 kB/s | 5.9 kB    00:00
CentOS Stream 9 - Extras packages                       5.5 kB/s | 21 kB    00:03
Dependencies resolved.
=====
Package                        Architecture      Version              Repository           Size
=====
Installing:
```

2. Create a directory to share using NFS:

#Mkdir /nfs_share

```
[root@server ~]# mkdir /nfs_share
[root@server ~]# ls
anaconda-ks.cfg
[root@server ~]# cd /
[root@server /]# ls
afs  bin  boot  dev  etc  home  lib  lib64  media  mnt  nfs_share  opt
[root@server /]#
```

#files in nfs_share

```
[root@server ~]# touch /nfs_share/file{1..10}.txt
[root@server ~]# cd /nfs_share/
[root@server nfs_share]# ls
file10.txt file1.txt file2.txt file3.txt file4.txt file5.txt file6.txt file7.txt file8.txt file9.txt
[root@server nfs_share]#
```

3. Edit the /etc/exports file to allow access to the directory:

#vi /etc/exports

```
[root@server ~]# cd /etc
[root@server ~]# vi /etc/exports
```

4. Add the lines in the configuration file to share the dir :

/nfs_share <server ip>(rw, sync)

```
/nfs_share      192.168.15.128(rw)
~
~
```

Wq! Save and exit

5. enable the service and To verify that the NFS server is running:

```
[root@server ~]# systemctl enable nfs-server.service --now
Created symlink /etc/systemd/system/multi-user.target.wants/nfs-server.service → /usr/lib/systemd/system/nfs-server.service.
[root@server ~]# systemctl status nfs-server.service
● nfs-server.service - NFS server and services
   Loaded: loaded (/usr/lib/systemd/system/nfs-server.service; enabled; preset: disabled)
   Drop-In: /run/systemd/generator/nfs-server.service.d
            └─order-with-mounts.conf
   Active: active (exited) since Fri 2026-06-26 11:32:22 IST; 22s ago
     Docs: man:rpc.nfsd(8)
           man:exportfs(8)
   Process: 33670 ExecStartPre=/usr/sbin/exportfs -r (code=exited, status=0/SUCCESS)
   Process: 33671 ExecStart=/usr/sbin/rpc.nfsd (code=exited, status=0/SUCCESS)
   Process: 33691 ExecStart=/bin/sh -c if systemctl -q is-active gssproxy; then systemctl reload gssproxy ; fi (code=exited
   Main PID: 33691 (code=exited, status=0/SUCCESS)
      CPU: 101ms

Jun 26 11:32:21 server systemd[1]: Starting NFS server and services...
Jun 26 11:32:22 server systemd[1]: Finished NFS server and services.
```

6.systemctl disable firewalld

7.check the getenforce -permissive

8.setenforce 0 then check the getenforce

9.reboot or init 6

Now go to client

1. Install the NFS client package

yum install nfs-utils

```
[root@client ~]# yum install nfs-utils
Last metadata expiration check: 0:11:42 ago on Fri 26 Jun 2026 11:27:12 AM IST.
Dependencies resolved.
=====
Package                               Architecture      Version            Repository          Size
=====
Installing:
nfs-utils                             x86_64            1:2.5.4-43.el9    baseos              463 k
Installing dependencies:
gssproxy                             x86_64            0.8.4-7.el9       baseos              110 k
libev                                 x86_64            4.33-6.el9        baseos              52 k
libnfsidmap                           x86_64            1:2.5.4-43.el9    baseos              62 k
libverto-libev                        x86_64            0.3.2-3.el9       baseos              14 k
rpcbind                               x86_64            1.2.6-7.el9       baseos              58 k
sssd-nfs-idmap                        x86_64            2.9.9-1.el9       baseos              37 k
=====
```

2. Create a directory to mount the NFS share:

mkdir /mnt/nfs_share

```
Complete:
[root@client ~]# mkdir /mnt/nfs_share
[root@client ~]# cd /mnt
[root@client mnt]# ls
hgfs  nfs_share
```

3. Mount the NFS share in the client's dir:

mount <nfs_server_ip>:/nfs_share /mnt/nfs_share

```
[root@client ~]# mount 192.168.15.130:/nfs_share /mnt/nfs_share
mount.nfs: No route to host
[root@client ~]# mount 192.168.15.130:/nfs_share /mnt/nfs_share
```

4. Verify that the NFS share has mounted:

df -h

```
[root@client ~]# df -h
Filesystem                Size      Used Avail Use% Mounted on
devtmpfs                   1.3G         0  1.3G   0% /dev
tmpfs                      1.3G         0  1.3G   0% /dev/shm
tmpfs                      522M     9.5M  513M   2% /run
/dev/mapper/cs-root        27G       5.4G   21G  21% /
/dev/sda1                  960M     562M  399M  59% /boot
tmpfs                      261M    108K   261M   1% /run/user/1000
/dev/sr1                   15G       15G     0 100% /run/media/raj/CentOS-Stream-9-BaseOS
tmpfs                      261M     52K   261M   1% /run/user/42
192.168.15.130:/nfs_share  27G       6.3G   20G  25% /mnt/nfs_share
```

6.I want permanent means

Using the command

#vi /etc/fstab

Edit : 192.168.15.130:/nfs_share /mnt/nfs_share nfs default 0 0

5.list

```
[root@client /]# cd /mnt/nfs_share
[root@client nfs_share]# ls
file10.txt file1.txt file2.txt file3.txt file4.txt file5.txt file6.txt file7.txt file8.txt file9.txt
[root@client nfs_share]#
```

6.create

```
[root@server /]# cd
[root@server ~]# chmod 777 /nfs_share/
```

```
mkdir: cannot create directory 'nfs': permission denied
[root@client nfs_share]# touch sky
[root@client nfs_share]# ls
file10.txt file1.txt file2.txt file3.txt file4.txt file5.txt file6.txt file7.txt file8.txt file9.txt sky
[root@client nfs_share]# ls -lh
total 0
-rw-r--r--. 1 root root 0 Jun 26 11:36 file10.txt
-rw-r--r--. 1 root root 0 Jun 26 11:36 file1.txt
-rw-r--r--. 1 root root 0 Jun 26 11:36 file2.txt
-rw-r--r--. 1 root root 0 Jun 26 11:36 file3.txt
-rw-r--r--. 1 root root 0 Jun 26 11:36 file4.txt
-rw-r--r--. 1 root root 0 Jun 26 11:36 file5.txt
-rw-r--r--. 1 root root 0 Jun 26 11:36 file6.txt
-rw-r--r--. 1 root root 0 Jun 26 11:36 file7.txt
-rw-r--r--. 1 root root 0 Jun 26 11:36 file8.txt
-rw-r--r--. 1 root root 0 Jun 26 11:36 file9.txt
-rw-r--r--. 1 nobody nobody 0 Jun 26 12:31 sky
[root@client nfs_share]#
```

How to change nobody nobody

#vi /etc/exports

```
/nfs_share 192.168.15.128(rw,no_root_squash)
```

Reboot or us this command exports -rav

```
[root@server ~]# exportfs -rav  
exporting 192.168.15.128:/nfs_share
```

Go to client

Create the file

```
-rw-r--r--. 1 nobody nobody 0 Jun 26 12:31 sky  
[root@client nfs_share]# touch ls  
[root@client nfs_share]# ls  
file10.txt file1.txt file2.txt file3.txt file4.txt file5.txt file6.txt file7.txt file8.txt file9.txt ls sky
```

List

```
[root@client nfs_share]# ls -lh  
total 0  
-rw-r--r--. 1 root root 0 Jun 26 11:36 file10.txt  
-rw-r--r--. 1 root root 0 Jun 26 11:36 file1.txt  
-rw-r--r--. 1 root root 0 Jun 26 11:36 file2.txt  
-rw-r--r--. 1 root root 0 Jun 26 11:36 file3.txt  
-rw-r--r--. 1 root root 0 Jun 26 11:36 file4.txt  
-rw-r--r--. 1 root root 0 Jun 26 11:36 file5.txt  
-rw-r--r--. 1 root root 0 Jun 26 11:36 file6.txt  
-rw-r--r--. 1 root root 0 Jun 26 11:36 file7.txt  
-rw-r--r--. 1 root root 0 Jun 26 11:36 file8.txt  
-rw-r--r--. 1 root root 0 Jun 26 11:36 file9.txt  
-rw-r--r--. 1 root root 0 Jun 26 12:44 ls  
-rw-r--r--. 1 nobody nobody 0 Jun 26 12:31 sky
```